

TranscendiantRouter

Every token earned. Most answers free.

AMD Developer Hackathon ACT II | Track 1: General-Purpose AI Agent

Team Transcendiant | solo build by David Castro

THE GAME

Scoring rewards the frugal, not the flashy

- Every answer is judged for accuracy. Below a hidden threshold, you are out.
- Everyone who passes is ranked by ONE thing: fewest Fireworks tokens.
- Local model tokens count as ZERO.
- So the winner is whoever knows exactly when NOT to pay.

THE INSIGHT

Free tokens can buy calibration

- A local model's confidence is a bad lie detector.
- But its CONSISTENCY is a good one.
- Ask it the same question several times. Agreement predicts correctness.
- Local sampling is free under the rules, so the lie detector costs nothing.

ARCHITECTURE

Classify, vote, then decide who answers

- 1. Rule-based classifier sorts each task into 8 categories. Zero tokens.
- 2. Qwen3-1.7B (llama.cpp, weights baked in) votes on each task.
- 3. Votes agree past a tuned threshold: ship the free answer.
- 4. Votes scatter: escalate that one task via Fireworks. Gemma 4 by default,
 - a code specialist for code.

ENGINEERING FOR THE GRADING BOX

4GB RAM. 2 vCPUs. 10 minutes. 10GB image.

- 3.1GB image with model weights baked in. Boots in seconds.
- Per-category token caps and adaptive vote counts sized for 2 CPU cores.
- A ratcheting time guard sheds sampling before the deadline is ever at risk.
- Dead remote model? Local fallback. A task can never return blank.
- Off-list model calls are impossible by construction.

MEASUREMENT

A 227-task benchmark harness, built first

- GSM8K math, HumanEval code, and authored tasks across all 8 categories.
- Held-out split never touched by tuning.
- Executable grading: generated code actually runs against test suites.
- Every routing threshold comes from recorded runs, not intuition.

THE LADDER

One recording, every strategy

- Record the local model once, record each expert once.
- Then replay every threshold combination offline, instantly.
- Result: a ladder from 77% accuracy at 0 tokens to 95% at 47K.
- Submissions walk down the ladder until the accuracy gate bites.

RESULTS

Dress rehearsal on a clone of the grading machine

- 92.5% accuracy on held-out tasks it had never seen.
- 5 minutes 58 seconds end to end, inside 4GB and 2 vCPUs.
- Most answers free. Escalations cost ~6.6K tokens per 40 tasks.
- Send-everything-remote baseline: ~50K+ tokens for similar accuracy.

GEMMA VIA FIREWORKS

The escalation brain, not a cameo

- Six of eight categories escalate to Gemma 4 31B through Fireworks.
- Model IDs resolve from ALLOWED_MODELS at runtime. Nothing hardcoded.
- Local Gemma was tested and ruled out honestly: the 4GB grading box
 - cannot hold the smallest Gemma 4 plus an agent.
- Gemma answers directly, with no hidden reasoning tokens to pay for.

Built solo, in four days, in public

- AMD GPU pod + GCP for development. llama.cpp + Qwen3 + Gemma + Fireworks.
- Reproducible: harness, tuning tools, and ladder ship in the repo.
- github.com/DavidOrtsac/frugal-router | MIT license